Global CO_2 Emissions: Trends, Regional Shifts, and Per Capita Inequalities

Scope: 1750–2024

A data-driven briefing for policy analysts and climate negotiators

Data sources: Global Carbon Budget 2025, NASA

Atmospheric CO_2 Reaches 429 ppm — 50% Above Pre-Industrial Levels

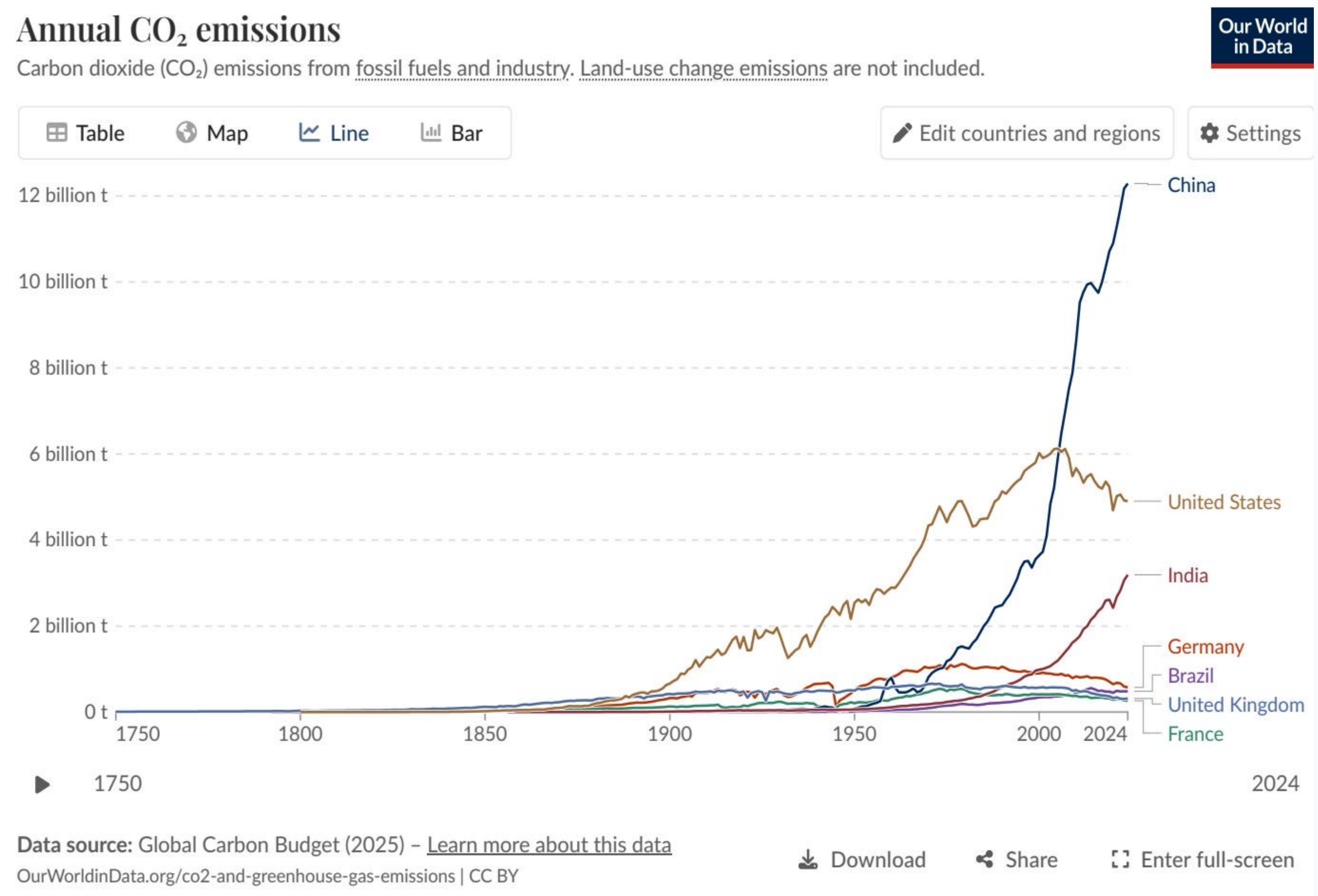
429 ppm

Measured at Mauna Loa
(February 2026)

- Pre-industrial baseline: ~280 ppm
- Increase: ~149 ppm (about +50%)
- Instrumental record began in 1958
- Acceleration visible over the modern era

Source: Global Carbon Budget 2025; NASA

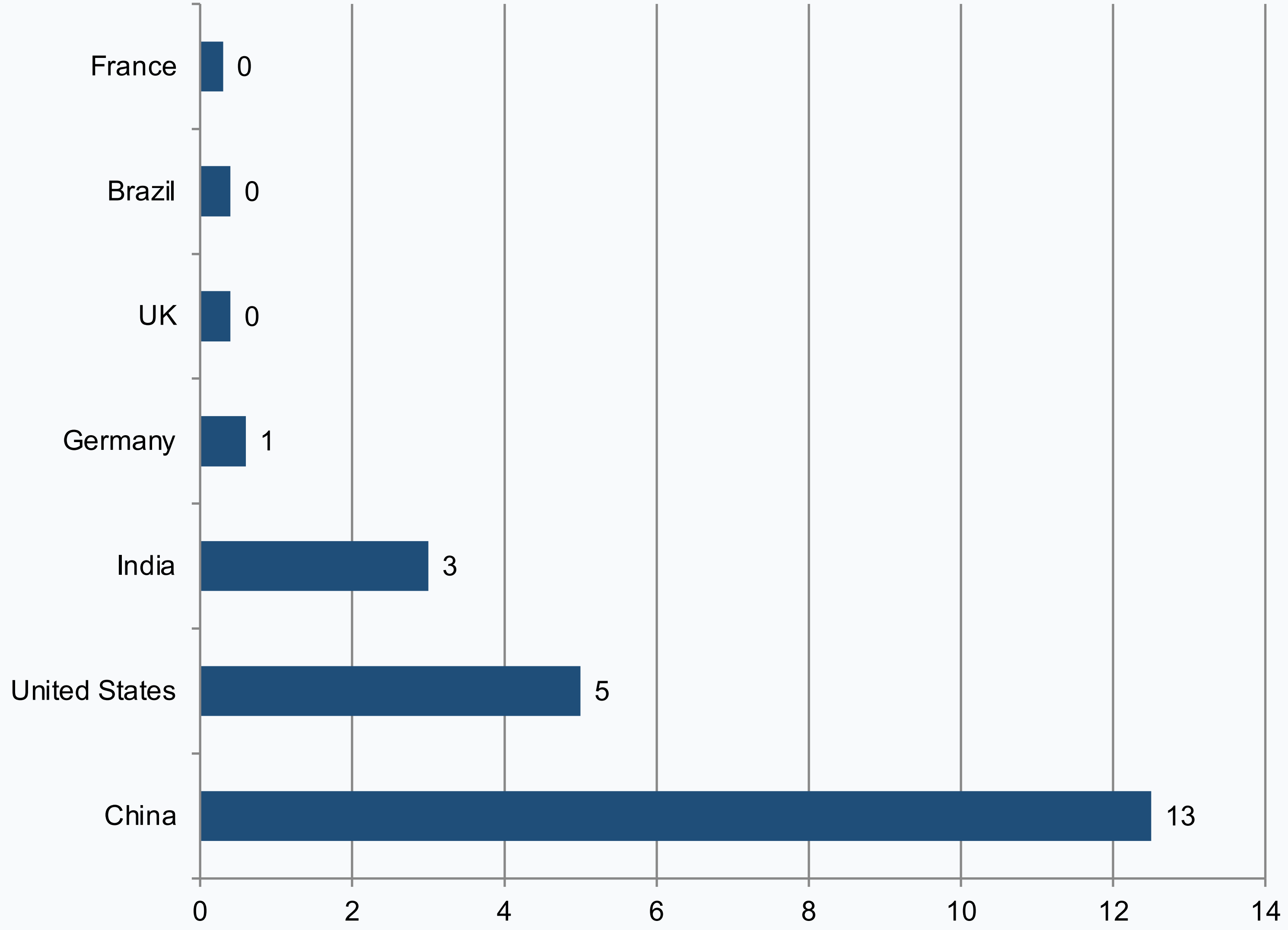
Global Fossil Fuel CO_2 Emissions Grew Six-Fold: 6 Billion to 35+ Billion Tonnes

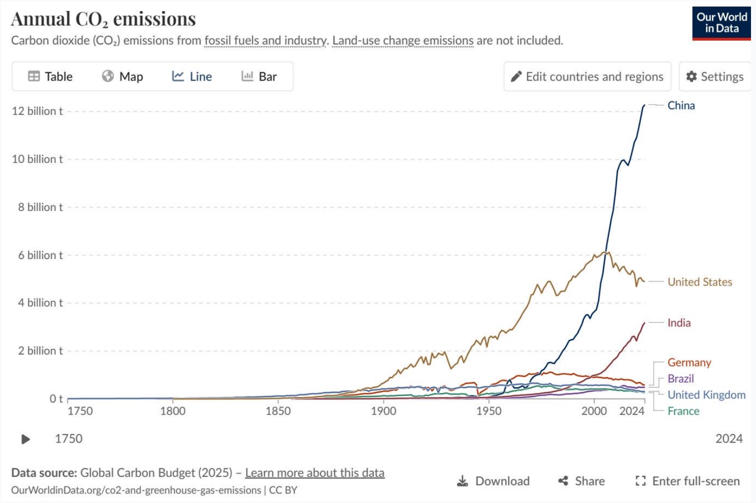


- ~6 Bt in 1950
- ~20 Bt by 1990
- >35 Bt by 2024

Source: Global Carbon Budget 2025; NASA

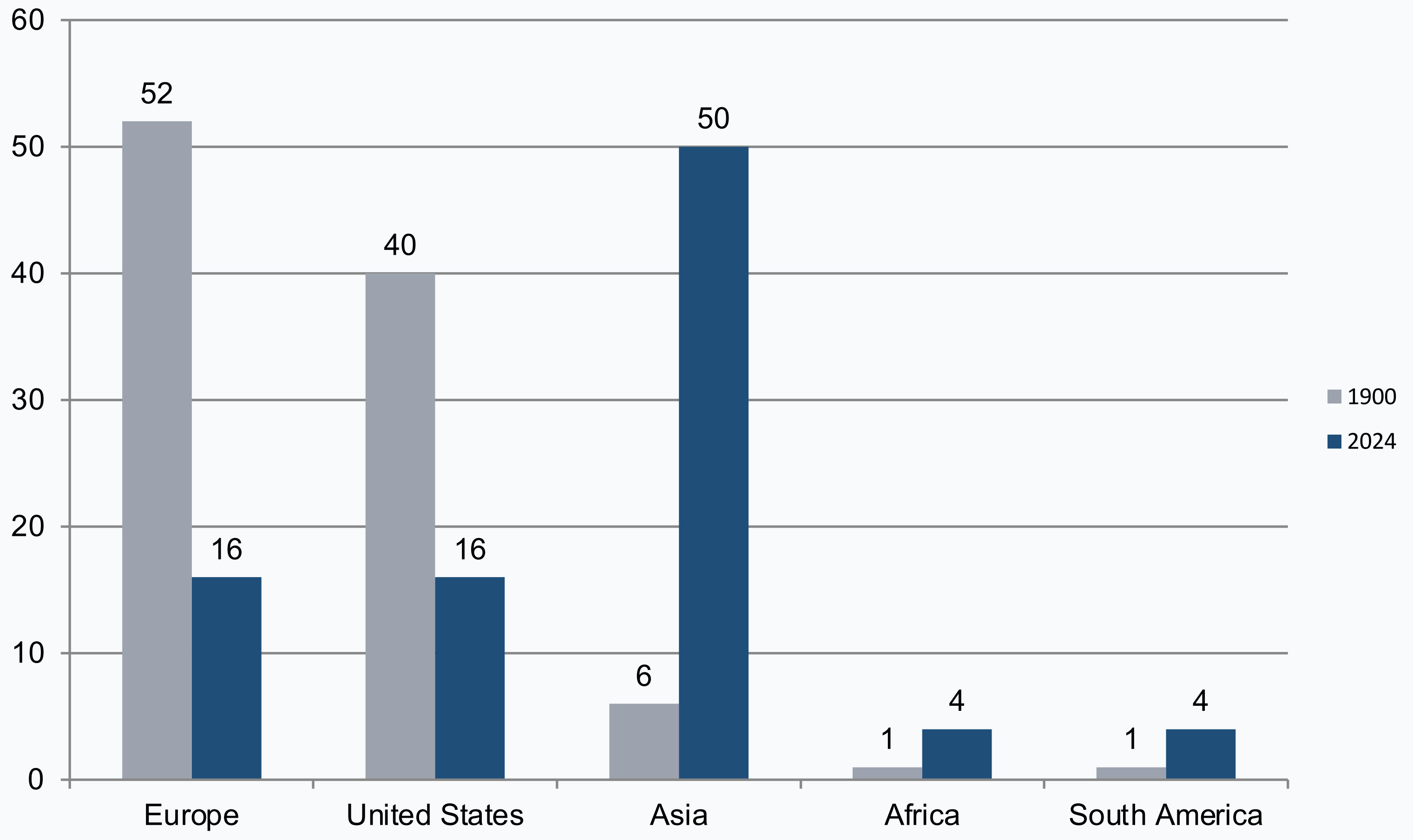
China Now Emits Over 12 Billion Tonnes — More Than Double the United States





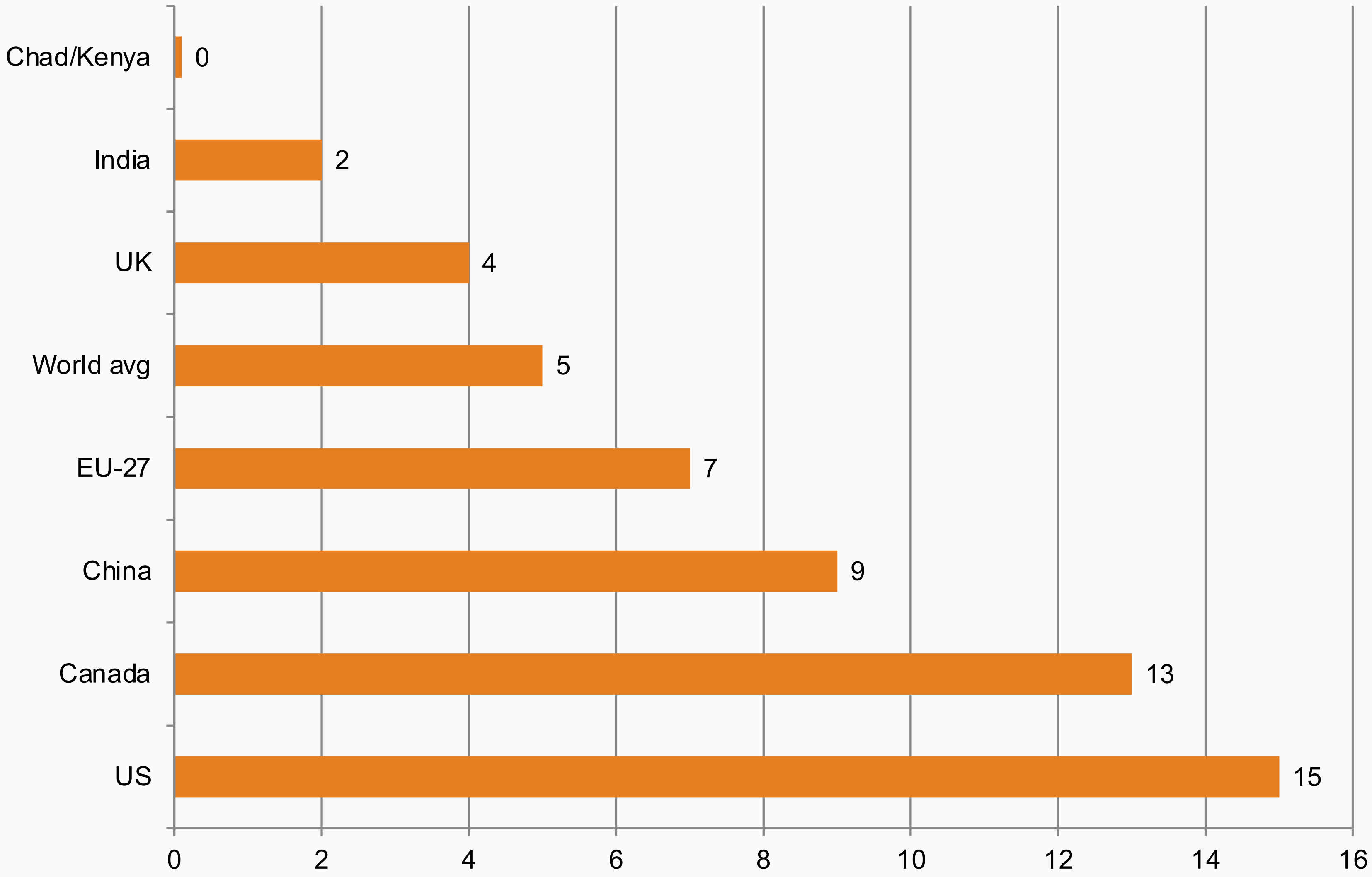
Source: Global Carbon Budget 2025; NASA

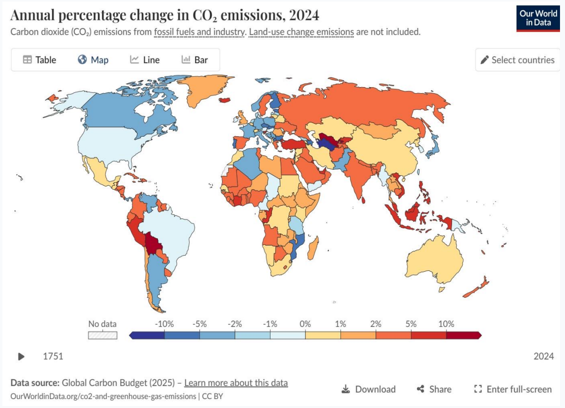
Asia Now Accounts for ~50% of Global Emissions; US + Europe Below One-Third



Source: Global Carbon Budget 2025; NASA

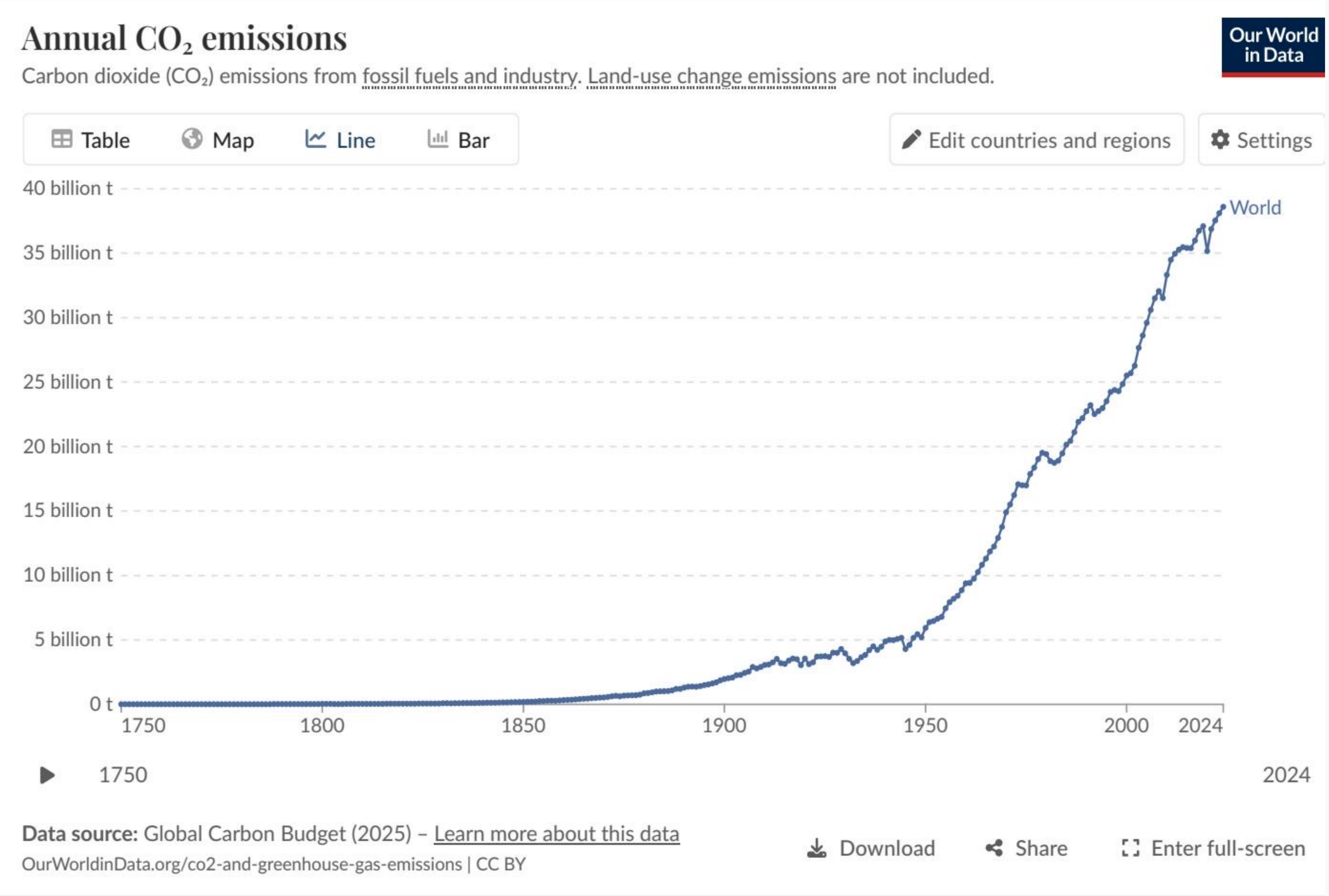
Per Capita Gap: 15 t in the US vs. 0.1 t in Chad — a 150× Disparity





Source: Global Carbon Budget 2025; NASA

2024 Snapshot: US and Europe Cut Emissions While Russia and SE Asia Increase



US/Europe declines (~−1% to −5%); Russia, SE Asia, and parts of Africa increase (+2% to +10% or more).

Source: Global Carbon Budget 2025; NASA

Energy Mix Explains Why France Emits Far Less Per Capita Than Germany

Dominant power mix	Higher nuclear + renewables	Higher fossil fuel share	Generation mix drives intensity
Per capita CO_2 signal	Lower among advanced economies	Higher than France/UK benchmark	System design can decouple prosperity from emissions

Key Takeaways: Emissions Are Still Rising, But the Map of Responsibility Is Shifting

1. Global fossil fuel CO_2 emissions exceed 35 billion tonnes and have not yet peaked.
2. China contributes over one-quarter of annual global emissions and emits more than double the United States.
3. Asia is near 50% of global emissions; US + Europe are below one-third.
4. Per capita inequality remains stark (~150× gap).
5. Energy-system choices materially shape per capita outcomes.